09.

Ratio & Proportion

Type - 8

1. Find the ratio of 3 days with 30 hours.
(a) 7 : 6 (b) 12 : 5
(c) 6 : 7 (d) 5 : 12
RRB RPF Constable -19/01/2019 (Shift-II)

Ans. (b) : $\because$ One day = 24 hours
$\therefore$ 3 days = 24 × 3 = 72 hours
Ratio = 72 : 30 = 12 : 5

2. Which of the following ratio is the largest?
(a) 5 : 7 (b) 2 : 3
(c) 3 : 4 (d) 3 : 5
RRB JE - 02/06/2019 (Shift-II)

Ans. (c) Largest ratio is
(a) 5/7 = 0.714
(b) 2/3 = 0.666
(c) 3/4 = 0.75
(d) 3/5 = 0.6
So option (c) is largest

3. A bag contained red, green and pink tokens. The ratio between red and green tokens was 15 : 32 while the ratio between pink and red tokens was 18 : 25. What was the ratio between green and pink tokens?
(a) 80 : 27 (b) 192 : 125
(c) 16 : 9 (d) 25 : 28
RRB RPF SI -05/01/2019 (Shift-II)

Ans : (a) Green : Red = 32 : 15
Red : Pink = 25 : 18
Green : Red : Pink = 800 : 375 : 270
So Green : Pink = 800 : 270 = 80 : 27

4. The ratio of the number of marbles that Tulip and Devansh had was 7 : 9 while Sheetal and Devansha had was 7 : 15. Find the ratio of number of marbles available with Tulip and Sheetal.
(a) 5 : 3 (b) 5 : 7
(c) 7 : 5 (d) 2 : 3
RRB RPF Constable -17/01/2019 (Shift-III)

Ans. (a) : $\frac{\text{Tulip}}{\text{Devansh}} = \frac{7}{9}$
and $\frac{\text{Sheetal}}{\text{Devansh}} = \frac{7}{15}$
so $\frac{\text{Tulip}}{\text{Sheetal}} = \frac{7}{9} \times \frac{15}{7} = \frac{5}{3}$

5. If a : b = 32 : 35 and b:c = 21 : 32. Then a : c ?
(a) 1 : 1 (b) 5 : 7
(c) 3 : 5 (d) 5 : 3
RRB Group-D – 03/10/2018 (Shift-I)

Ans : (c)
$\frac{a}{b} = \frac{32}{35}$ ..... (i)
$\frac{b}{c} = \frac{21}{32}$ ..... (ii)
From the equation (i) and (ii)
$\frac{a}{b} \times \frac{b}{c} = \frac{32}{35} \times \frac{21}{32}$
$\frac{a}{c} = \frac{3}{5}$

6. Two numbers are 30% and 60% higher than the third number respectively. What is the ratio of both the numbers ?
(a) 14 : 13 (b) 16 : 13
(c) 22 : 23 (d) 13 : 16
RRB Group-D – 10/10/2018 (Shift-II)

Ans : (d) Suppose third number = 100
$\therefore$ First number and second number = 130, 160
So required ratio $\frac{130}{160} = 13:16$

7. The ratio of sand and Macadam in a mixture is 41 : 30. While the mixture of Macadam and cement is 6 : 7. What is the ratio of sand and cement in the mixture?
(a) 8 : 6 (b) 11 : 7
(c) 77 : 48 (d) 41 : 35
RRB Group-D – 16/10/2018 (Shift-I)

Ans. (d)
$\frac{\text{Sand}}{\text{Macadam}} = \frac{41}{30}$ and $\frac{\text{Cement}}{\text{Macadam}} = \frac{7}{6} = \frac{7 \times 5}{6 \times 5} = \frac{35}{30}$
So $\frac{\text{Sand}}{\text{Cement}} = \frac{\text{Sand}}{\text{Macadam}} \times \frac{\text{Macadam}}{\text{Cement}} = \frac{41}{30} \times \frac{30}{35} = \frac{41}{35}$
$= 41:35$

8. Red, green and pink tokens are kept in a bag, the ratio of red and green is 5 : 11 while the ratio of pink and red tokens is 7 : 15. What will be the ratio of green and pink?
(a) 77 : 75 (b) 11 : 7
(c) 33 : 7 (d) 75 : 77
RRB Group-D – 25/09/2018 (Shift-I)

Ans : (c) Pink : Red = 7 : 15
Red : Green = 5 : 11
Then, Pink : Red : Green = 7 × 5 : 5 × 15 : 15 × 11
= 35 : 75 : 165
Green : Pink = 165 : 35 = 33 : 7

9. If A : B = 5 : 8 and B : C = 18 : 25 then find A : C.
(a) 8 : 5 (b) 9 : 20
(c) 5 : 8 (d) 20 : 9
RRB Group-D – 09/10/2018 (Shift-I)

Ans. (b) : Given

$$\frac{A}{B}=\frac{5}{8} \text{ and } \frac{B}{C}=\frac{18}{25}$$

$$\text{Then } \frac{A}{C}=\left(\frac{A}{B}\times\frac{B}{C}\right)=\left(\frac{5}{8}\times\frac{18}{25}\right)=\frac{90}{200}$$

So A : C = 9 : 20

10. The ratio of soil and gravel in a mixture is 11 : 8 while the ratio of gravel and cement is 6 : 7. What is the ratio of soil and cement in the mixture?
(a) 77 : 48 (b) 33 : 28
(c) 8 : 6 (d) 11 : 7
RRB Paramedical Exam – 20/07/2018 (Shift-I)

Ans : (b)
(i) Soil (S) gravel (B)
11 : 8
(ii) gravel (B) Cement (C)
6 : 7
S : B = 11 : 8 = 66 : 48
B : C = 6 : 7 = 48 : 56
S : B : C = 66 : 48 : 56
Soil : Cement = 66 : 56
Soil : Cement = 33 : 28

11. If the ratio of a : b is 45 : 56 and the ratio of b : c is 16 : 35, then what is the ratio of a : c?
(a) 9 : 7 (b) 18 : 49
(c) 7 : 2 (d) 7 : 9
RRB Group-D – 01/12/2018 (Shift-II)

Ans : (b) Given
a : b = 45 : 56
b : c = 16 : 35

$$\therefore a:c=\frac{a}{c}=\frac{a}{b}\times\frac{b}{c}=\frac{45}{56}\times\frac{16}{35}$$

$$=\frac{18}{49}=18:49$$

12. Suraj's amount is 4 times that of Ravi's amount. Ravi's amount is 16 times the amount of Aditya's amount. What is the ratio of Aditya's and Suraj's amount?
(a) 64 : 1 (b) 1 : 64
(c) 1 : 24 (d) 1 : 16
RRB Group-D – 11/12/2018 (Shift-II)

Ans : (b) According to the question
Let aditya's amount = Rs. x
and Ravi's amount = Rs. 16x
Suraj's amount = Rs. 64x
Ratio of amount of Aditya, Ravi and Suraj
= x : 16x : 64x
Ratio of amount of Aditya and Suraj
= x : 64x = 1 : 64

13. If a : b = 2 : 3 and a : c = 10 : 21, then what is b : c?
(a) 5 : 7 (b) 15 : 14
(c) 14 : 15 (d) 7 : 5
RRB Group-D – 26/10/2018 (Shift-II)

Ans : (a) Given

$$\frac{a}{b}=\frac{2}{3},\quad \frac{a}{c}=\frac{10}{21}$$

$$\frac{c}{b}=\frac{a}{b}\times\frac{c}{a}=\frac{2}{3}\times\frac{21}{10}$$

c: b = 7 : 5
b : c = 5 : 7

14. If L:M = 3:5 and M:N = 2:3, then N:L = ?
(a) 2:1 (b) 5:2
(c) 3:2 (d) 1:2
RRB NTPC 29.03.2016 Shift : 3

Ans : (b) L : M = 3 : 5
M : N = 2 : 3
By equating the ratio M
L : M = 6 : 10
M : N = 10 : 15
∴ L : M : N = 6 : 10 : 15
So N : L = 15 : 6 = 5 : 2

15. If A : B = 2 : 5 and B : C = 3 : 4, then A : C = ?
(a) 1:2 (b) 3:10
(c) 2:3 (d) 5:4
RRB NTPC 28.03.2016 Shift : 2

Ans :(b) A : B = 2:5.....(i)
B : C = 3:4.....(ii)
Multiplying to 3 in equation (i) and 5 in (ii)
A : B = 6:15
B : C = 15:20
Required ratio A : B : C = 6:15:20
So A : C = 6:20
A : C = 3:10

16. Two numbers are 40% and 60% more than the third number respectively, what is the ratio of the first number to the second number?
(a) 7 : 9 (b) 8 : 9
(c) 8 : 7 (d) 7 : 8
RRB NTPC 18.01.2017 Shift : 3

Ans : (d) Let the third number = 100
I — 140
II — 160
III — 100

$$\Rightarrow \frac{I_{num}}{II_{num}}=\frac{140}{160}=\boxed{7:8}$$

17. If A : B = 3 : 4 and B : C = 6 : 5, then A : (A + C) = ?
(a) 9:11 (b) 9:10
(c) 9:19 (d) 6:7
RRB NTPC 16.04.2016 Shift : 3

Ans : (c)

A : B : C

3 4

6 5

$\Rightarrow$ 18 : 24 : 20

$\Rightarrow$ 9 : 12 : 10

अत: $\frac{A}{A+C} = \frac{9}{9+10} = \frac{9}{19}$

18. If a/b = 1/4, b/c = 18 and a = 2 then the value of c is-

(a) 8 (b) 16

(c) 32 (d) 64

RRB NTPC 27.04.2016 Shift : 3

Ans : (d) $\frac{a}{b} = \frac{1}{4} \Rightarrow \frac{2}{b} = \frac{1}{4}$ $(\because a = 2)$

$b = 2 \times 4$

$b = 8$

and $\frac{b}{c} = \frac{1}{8} \Rightarrow \frac{8}{c} = \frac{1}{8}$

$c = 8 \times 8 \Rightarrow c = 64$

19. The ratio of the number of marbles that Joyee and Minit had was 5 : 8 while the ratio of the number of marbles that Jacob and Minit had was 7 : 12. What is the ratio of the number of marbles that Joyee and Jacob had ?

(a) 7 : 5 (b) 2 : 3

(c) 15 : 14 (d) 5 : 7

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (c) Joyee ← Minit ← Jacob

5 8 7

12

60	96	56
30	48	28
15	24	14

Hence ratio of Joe and Jacob's marbles = 15:14

20. The ratio of sand to gravel in a mixture is 7:8 while that between gravel and cement is 6:7. What is the ratio of sand to cement in the mixture?

(a) 49 : 48 (b) 7 : 7

(c) 8 : 6 (d) 3 : 4

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (d)

$\therefore \frac{\text{Sand}}{\text{Gravel}} = \frac{7}{8}$ and $\frac{\text{Gravel}}{\text{Cement}} = \frac{6}{7}$

$\therefore \frac{\text{Sand}}{\text{Cement}} = \frac{\text{Sand}}{\text{Gravel}} \times \frac{\text{Gravel}}{\text{Cement}}$

$= \frac{7}{8} \times \frac{6}{7} = \frac{6}{8} = 3:4$

21. The ratio 1 : 5 can be expressed in decimal as:

(a) 0.5 (b) 0.2

(c) 0.02 (d) 0.1

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (b) $1:5 = \frac{1}{5} = 0.2$

22. In a bag containing red, green and pink tokens, the ratio of red to green tokens was 5 :12 while the ratio of pink to red tokens was 7 : 15. What was the ratio of green to pink tokens ?

(a) 25 : 28 (b) 36 : 7

(c) 28 : 25 (d) 12 : 7

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (b) Given

Green tokens : Reds tokens =12 : 5

Red tokens : Pink tokens =15 : 7

Then, green tokens : red tokens : pink tokens = 12×15 : 15×5 : 7×5

180 : 75 : 35

green tokens : pink tokens

180 : 35 = 36 : 7

23. The ratio of sand to gravel in a mixture is 17 : 8 while that between gravel and cement is 6 : 17. What is the ratio of sand to cement in the mixture?

(a) 17 : 17 (b) 289 : 48

(c) 8 : 6 (d) 3 : 4

RRB ALP & Tec. (09-08-18 Shift-III)

Ans : (d) R : B = 17 : 8, B : S = 6 : 17

R : B : S = 102 : 48 : 136

Then ratio of sand (R) and cement (S) mixture

= 102 : 136 = 3 : 4

24. If a : b = $\frac{3}{2} : \frac{7}{3}$ and b : c = $\frac{1}{5} : \frac{1}{7}$ Then find a : b : c ?

(a) 14:9:10 (b) 4:5:7

(c) 9:14:10 (d) 10:9:14

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (c) $a : b = \frac{3}{2} : \frac{7}{3} = 9:14$

$b : c = \frac{1}{5} : \frac{1}{7} = (7:5) \times 2 = 14:10$

$\therefore a : b : c = 9 : 14 : 10$

25. a : b = 7 : 9, b : c = 5 : 11. Then find a : b : c ?

(a) none of these (b) 99 : 45 : 35

(c) 45 : 35 : 99 (d) 35 : 45 : 99

RRB Group-D – 11/10/2018 (Shift-III)

Ans : (d)

a : b = 7 : 9

b : c = 5 : 11

Hence a : b : c = 35 : 45 : 99

26. If a : b = 3 : 5 , c : b = 3 : 2 , c : d = 5 : 6 then find a : d = ?

(a) 12 : 36 (b) 12 : 15

(c) 1 : 3 (d) 11 : 36

RRB NTPC 18.04.2016 Shift : 3

Ans : (c) a : b = 3 : 5, c : b = 3 : 2, c : d = 5 : 6

$\Rightarrow$ b : c = 2 : 3

$\therefore a : d = \frac{a}{b} \times \frac{b}{c} \times \frac{c}{d}$

$= \frac{3}{5} \times \frac{2}{3} \times \frac{5}{6} = \frac{1}{3}$

Type - 2

27. On dividing of Rs. 3900 in between L, K and J by the ratio of $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$, what amount will k get?

(a) Rs. 1,450 (b) Rs. 30
(c) Rs. 1,200 (d) Rs. 900

RRB RPF Constable -18/01/2019 (Shift-III)

Ans : (c) Ratio between L, K and J $= \frac{1}{2} : \frac{1}{3} : \frac{1}{4}$

$= 6 : 4 : 3$

So received amount of K $= 3900 \times \frac{4}{13}$ = Rs. 1200

29. In an alloy of German Silver the ratio of Copper and Zinc was 21 : 16, while the ratio of Nickel and Zink was 7 : 24. What was the ratio of Copper, Zink and Nickel in the alloy?

(a) 63:48:14 (b) 21:6:7
(c) 17:21:4 (d) 68:28:21

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (a) : Copper : Zinc = 21 : 16

Nickel : Zinc = 7 : 24

Copper : Zinc : Nickel $= 21 \times 24 : 16 \times 24 : 7 \times 16$

$= 3 \times 21 : 3 \times 16 : 2 \times 7$

$= 63 : 48 : 14$

30. In an alloy of German Silver the ratio of Copper and Zinc was 19 : 6 while the ratio of Nickel and Zink was 7:4. Then what was the ratio of Copper, Zink and Nickel?

(a) 19 : 44 : 4 (b) 19 : 24 : 7
(c) 38 : 12 : 21 (d) 133 : 42 : 24

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (c) Copper : Zinc = 19 : 6

Nickel : Zinc = 7 : 4

Copper : Zinc : Nickel = 19 : 6
4 : 7

= 76 : 24 : 42

= 38 : 12 : 21

31. In an alloy of German silver the ratio of copper to zinc was 17 : 7 while that of nickel to zinc was 4 : 3. The ratio of copper to zinc to nickel in the alloy was:

(a) 17 : 21 : 4 (b) 51 : 21 : 28
(c) 68 : 28 : 21 (d) 17 : 28 : 3

RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (b) C : Z = 17 : 7, C : Z = 4 : 3

or Z : N = 3 : 4

$\boxed{T : Z : C = 51 : 21 : 28}$

32. The ratio of Copper, Zinc and Nickel in German Silver is 4:3:2. How many kilograms of Zinc should be added to this metal of 54 kg so that the new ratio becomes 2:5:1.

(a) 50 (b) 48
(c) 36 (d) 42

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (d) Copper : Zinc : Nickel = 4 : 3 : 2

Suppose quantity is 4x, 3x and 2x kg respectively

According to the question

4x + 3x + 2x = 54

9x = 54

x = 6

$\therefore$ Copper = 4 × 6 = 24 kg

Zinc = 3 × 6 = 18 kg

Nickel = 2 × 6 = 12 kg

Let the new ratio will be 2:5:1 by adding y kg of Zinc

$\frac{24}{18+y} = \frac{2}{5}$

120 = 36 +2y

84 = 2y

y = 42 kg

So adding zinc in new metal = 42 kg

33. If a = 2b/3, b = 2c/3, and c = 2d / 30 what is the ratio of b and d ?

(a) 8/9 (b) 4/9
(c) 4/3 (d) 5/27

RRB NTPC 19.04.2016 Shift : 2

Ans : (b) Given

$a = \frac{2b}{3}, b = \frac{2c}{3}, c = \frac{2d}{3}$

$\Rightarrow \frac{a}{b} = \frac{2}{3}, \frac{b}{c} = \frac{2}{3}, \frac{c}{d} = \frac{2}{3}$

$\Rightarrow \frac{a}{b} = \frac{8}{12}, \frac{b}{c} = \frac{12}{18}, \frac{c}{d} = \frac{18}{27}$

$\Rightarrow$ a : b : c : d = 8 : 12 : 18 : 27

$\therefore \frac{b}{d} = \frac{12}{27}$ $\Rightarrow \boxed{\frac{b}{d} = \frac{4}{9}}$

34. The ratio of marks obtained by a student in three subject is 1 : 2 : 3 . The school has decided to give 5% grace marks for each subject. Find the new student ratio.

(a) 1:2:3 (b) 2:3:4
(c) 2:3:1 (d) 3:2:1

RRB NTPC 06.04.2016 Shift : 1

Ans : (a) Suppose mark score by students in 3 subject is x, 2x and 3x

According to the question

$x \times \frac{100+5}{100} : 2x \times \frac{100+5}{100} : 3x \times \frac{100+5}{100}$

x : 2x : 3x

1:2:3

35. Divide Rs. 169 in the ratio 2 : 5 : 6 what will be their share in rupees according to the above ratio?

(a) 26, 66, 77 (b) 26, 65, 78
(c) 25, 67, 78 (d) 26, 70, 73

RRB NTPC 19.01.2017 Shift : 3

Ans : (b) $\frac{169\times 2}{13}=26$

$\frac{169\times 5}{13}=65$

$\frac{169\times 6}{13}=78$

Split amount = 26, 65, 78

Type - 3

36. The ratio of two positive integer is 3 : 4. Their sum is 70, how much should be added to each integer so that their ratio become 5 : 6 ?

(a) 10 (b) 20
(c) 30 (d) 40

RRB RPF SI -12/01/2019 (Shift-III)

Ans : (b) First integer $=\frac{3}{7}\times 70=30$

Second integer $=\frac{4}{7}\times 70=40$

Suppose adding x in each integer ratio will be 5:6

Hence

$\frac{30+x}{40+x}=\frac{5}{6}$

$\Rightarrow 180 + 6x = 200 + 5x$

$\Rightarrow 6x - 5x = 200 - 180 \Rightarrow x = 20$

37. The ratio of two number is 3 : 5, If each number is increased by 10, the ratio become 5 : 7 find the smallest number?

(a) 8 (b) 12
(c) 15 (d) 18

RRB RPF SI -11/01/2019 (Shift-I)

Ans : (c) Let numbers are 3x and 5x

$\therefore \frac{3x+10}{5x+10}=\frac{5}{7}$

21x + 70 = 25x + 50

4x = 20 or x = 5

$\therefore$ Small number = 3x= 3 × 5 = 15

38. The salary of Charan and Rajat is in the ratio of 5 : 4 If the salary of each is increased by 3,000 then their new ratio become 6 : 5. What is salary of Charan.

(a) Rs. 15,000 (b) Rs. 12,000
(c) Rs. 8,000 (d) Rs. 20,000

RRB RPF Constable -20/01/2019 (Shift-I)

Ans : (a) Charan : Rajat
5x : 4x

According to the question

$\frac{5x+3000}{4x+3000}=\frac{6}{5}$

25x + 15000 = 24x + 18000

25x – 24x = 18000 – 15000

x = 3000

Salary of Charan = 5x = 5 × 3000

= Rs. 15000

39. The ratio of two number are 5 : 9. If 6 is added in both numbers then their ratio become 2 : 3. The original number are.

(a) 25, 45 (b) 10, 18
(c) 15, 27 (d) 5, 9

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (b) : Suppose both numbers are 5x and 9x respectively.

According to the question

$\frac{5x+6}{9x+6}=\frac{2}{3}$

$15x+18=18x+12$

$6=3x$

$x=2$

Original numbers = 5 × 2 and 9 × 2

= 10 and 18

40. A number is divided into a ratio of 7 : 12 when 6 is added to each number, the ratio is changed to 5 : 7. What was the larger number among the initial number.

(a) 22 (b) 11
(c) 33 (d) 44

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (a) Suppose numbers are 7x and 11x respectively. Then

$\frac{7x+6}{11x+6}=\frac{5}{7}$

49x + 42 = 55x + 30

42 – 30 = 55x – 49x

$12 = 6x \Rightarrow x = \frac{12}{6}=2$

$\boxed{x = 2}$

So numbers

7x = 7 × 2 = 14

11x = 11 × 2 = 22

Hence large number = 22

41. The ratio between two number are 11 : 18. If 4 is added to both numbers, then the ratio between them becomes 13 : 20 what will be original number.

(a) 21, 36 (b) 26, 21
(c) 22, 36 (d) 32, 23

RRB Group-D – 05/10/2018 (Shift-II)

Ans : (c) Let two number is 11x and 18x

Then $\frac{11x+4}{18x+4}=\frac{13}{20}$

220x + 80 = 234x + 52

28 = 14x

x = 2

Putting x=2 in 11x and 18x, respectively we get the numbers 22 and 36

42. The ratio of two number is 3 : 4. If 3 is added in each of them, then their ratio become 10 : 13. Find the original number.

(a) 9, 12 (b) 12, 16

(c) 20, 25 (d) 27, 36

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (d) Let the number is 3x and 4x

According to the question

$\frac{3x+3}{4x+3}=\frac{10}{13}$

$39x + 39 = 40x + 30$

$x = 9$

Original number $= 3x, 4x$

$= 3 \times 9, 4 \times 9$

$= \boxed{27, 36}$

43. The initial ratio of Sugar and Flour in a dough was 4 : 7 Justin Added more flour in that 22 kg dough change to the ratio of sugar and flour 2:5 how much flour did Justin later add?

(a) 2 kg (b) 4 kg

(c) 8 kg (d) 6 kg

RRB Group-D – 22/09/2018 (Shift-III)

Ans. (d) : Quantity of flour in dough of 22kg

$=22\times\frac{7}{11}=14\text{kg}$

Quantity of sugar = 22 – 14 = 8 kg

Let mixed quantity of flour = x kg

According to the question

$\frac{8}{14+x}=\frac{2}{5}$

40 = 28 + 2x

2x = 12

x = 6

Hence mixed quantity of flour = 6 kg

44. A number is divided in the ratio of 9 : 5 when 8 is added to each number, the ratio becomes 5 : 3 which will be the largest number among the two?

(a) 80 (b) 72

(c) 69 (d) 81

RRB Group-D – 25/09/2018 (Shift-II)

Ans : (b) Let the number is 9x, 5x

According to the question

$\frac{9x+8}{5x+8}=\frac{5}{3}$

27x + 24 = 25x + 40

2x = 40 – 24

2x = 16

x = 8

So large number = 9 × 8 = 72

45. The ratio of red ball and green ball in a bag is 15 : 26. If 12 more green ball are put in the bag then the ratio of red ball and green ball become 1 : 2. How many red ball in the bag?

(a) 60 (b) 30

(c) 45 (d) 15

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (c) : Red : Green = 15:26

Suppose red ball = 15x and green ball = 26x

Putting more 12 green ball

$\frac{15x}{26x+12}=\frac{1}{2}$

30x = 26x + 12

4x = 12

x= 3

Red ball in bag = 15×3 = 45

46. An amount was divided between Ethen and Jen in the ratio of 4 : 7. If Jen gives Ethen 1 rupee, then the ratio changes to 7 : 12. What is the total amount?

(a) Rs. 209 (b) Rs. 190

(c) Rs. 198 (d) Rs. 220

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (a) : Suppose amount of Ethen and Jen = 4x, 7x

Ratio after giving 1 Rs. to Ethen $=4x+1:7x-1=7:12$

$\Rightarrow 48x + 12 = 49x - 7$

$\Rightarrow x = 19$

Amount of Ethen = 4x = 4 × 19 = Rs. 76

Amount of Jen = 7x = 7 × 19 = Rs. 133

Total amount = 76 + 133 = Rs. 209

47. The ratio of two number are is 17 : 28, if 6 is added to smaller number, then the ratio changes to 13 : 20. What is the value of larger number?

(a) 112 (b) 140

(c) 98 (d) 126

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (b) Let the first number = 17x

Second number = 28x

According to the question

$\frac{17x+6}{28x}=\frac{13}{20}$

$\frac{17x+6}{7x}=\frac{13}{5}$

85x + 30 = 91x

6x = 30

x = 5

So large number = 28x = 28 × 5 = 140

48. Any amount is divided between Shivani and Parineeta in the ratio of 5 : 7, if Parineeta gives Rs. 5 to Shivani, then the ratio will be changed to 3 : 4 what is divided amount?

(a) Rs. 432 (b) Rs. 420

(c) Rs. 396 (d) Rs. 408

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (b) : Let distributed amount between Shivani and Parinita = 5x and 7x
According to the question,
$\frac{5x+5}{7x-5}=\frac{3}{4}$
$\Rightarrow 20x + 20 = 21x - 15$
x = 35
Splited amount = 5x + 7x = 12x
= 12 × 35 = Rs. 420

49. The ratio of two number are 2 : 3. If 12 is subtracted from both numbers then the ratio becomes 5 : 8. Find the number.
(a) 16 and 24 (b) 35 and 56
(c) 72 and 108 (d) 20 and 48
RRB NTPC 04.04.2016 Shift : 1

Ans : (c) Let number are 2x and 3x
$\frac{2x-12}{3x-12}=\frac{5}{8}$
16x – 96 = 15x – 60
x = 36
∴Numbers will be 72 and 108

50. The ratio of two numbers are 7:12. If 7 is added in both numbers then ratio becomes 7:11. Find the smallest number.
(a) 7 (b) 28
(c) 35 (d) 12
RRB NTPC 29.03.2016 Shift : 2

Ans : (b) Suppose small number and large number is 7x and 12x respectively.
From the question $\frac{7x+7}{12x+7}=\frac{7}{11}$
$\Rightarrow 77x + 77 = 84x + 49$
$7x = 28 \Rightarrow x = 4$
∴ Small number = 7x = 7 × 4 = 28

51. The ratio of boys and girls in the class are 4:5. If 4 new boys are included in class then the number of boys increases by 20%. Find the number of girls in the class.
(a) 30 (b) 35
(c) 20 (d) 25
RRB NTPC 18.01.2017 Shift : 3

Ans : (d) Let the number of boys = 4x
and the number of girls = 5x
According to the question
$\Rightarrow 4x+4=\frac{24x}{5}$
20x + 20 = 24x
20 = 4x $\boxed{x=5}$
Number of girls = 5x = 5 × 5 = 25

52. In a mixture of 25 litre, the ratio of Milk and Water is 4:1. Howmuch litre Milk should be added more so that the ratio becomes 16:1?
(a) 21 (b) 25
(c) 60 (d) 36
RRB NTPC 19.04.2016 Shift : 2

Ans : (c) Amount of milk in mixture of 25 litre
$=25\times\frac{4}{5}=20\,l$
Amount of water = 25–20 = 5*l*
Let amount of added milk is x*l*, According to the question
$\frac{20+x}{5}=\frac{16}{1}$ $\Rightarrow 20 + x = 16 \times 5$
$\Rightarrow x = 80–20$ $\Rightarrow x = 60$ liters

53. The ratio of two numbers are 5:6. When 6 is added to both numbers the ratio becomes 7:8 then the numbers are.
(a) 10, 12 (b) 20, 24
(c) 15, 18 (d) 5, 6
RRB NTPC 18.04.2016 Shift : 1

Ans : (c) Let numbers are 5x and 6x respectively
According to the question $\frac{5x+6}{6x+6}=\frac{7}{8}$
40x + 48 =42x + 42
2x = 6 x = 3
∴ Both numbers = 5x, 6x
= 5×3 and 6×3 = 15 and 18

54. The ratio of girls and boys in 504 students of a school are 11:13. If given admission to 4 more girls then find the new ratio.
(a) 31:51 (b) 91:81
(c) 81:91 (d) 51:31
RRB NTPC 18.04.2016 Shift : 2

Ans : (c) Total students = 504
ratio = 11:13
girls $=\frac{11}{24}\times 504$ = 231
boys = 504 − 231= 273
on adding 12 more girls
231 + 12 = 243
Required ratio = 243 : 273 = 81: 91

55. Before 4 years, the ratio of Vikas and Rahul was 3:5. After 6 years, this ratio will become 4:5. Find the present age of Rahul.
(a) 10 (b) 15
(c) 14 (d) 17
RRB NTPC 12.04.2016 Shift : 3

Ans : (c) Let present age of Vikas and Rahul is x and y respectively.
From the question,
$\frac{x-4}{y-4}=\frac{3}{5}$
$\Rightarrow 5x - 20 = 3y - 12$
$\Rightarrow 5x - 3y = 8$ ……(i)
again
$\frac{x+6}{y+6}=\frac{4}{5}$
$\Rightarrow 5x + 30 = 4y + 24$

$\Rightarrow 5x - 4y = -6$ .....(ii)
now from equation (i) - (ii)
$5x - 3y - (5x - 4y) = 8 - (-6)$
$\Rightarrow 5x - 3y - 5x + 4y = 8 + 6$
$\Rightarrow y = 14$
Hence present age of Rahul = 14 years

56. The ratio of two numbers are 4 : 5. If 5 is subtracted from both, then the new ratio becomes 3 : 4. Find the largest number.
(a) 30 (b) 25
(c) 20 (d) 15
RRB NTPC 26.04.2016 Shift : 1

Ans : (b) Let numbers are 4x and 5x
Subtracting 5 from both numbers
$\frac{4x-5}{5x-5} = \frac{3}{4}$
$\Rightarrow 16x - 20 = 15x - 15$
$\Rightarrow 16x - 15x = -15 + 20$
$x = 5$
Large number = $5x = 5 \times 5 = 25$

57. The ratio of two numbers are 3 : 4. When 2 is subtracted from both the numbers, the ratio becomes 2 : 3. Find the sum of the numbers.
(a) 16 (b) 20
(c) 21 (d) 22
RRB NTPC 27.04.2016 Shift : 3

Ans : (c) Let numbers are 3x and 4x
According to the question,
$\frac{3x-3}{4x-3} = \frac{2}{3}$
$\Rightarrow 9x - 9 = 8x - 6$
$\Rightarrow 9x - 8x = 9 - 6$
$x = 3$
$\therefore$ Sum of the numbers = $3x + 4x = 7x = 7 \times 3 = 21$

58. A number is divided in the ratio 3 : 2. When 8 is added to each of the numbers then the ratio changes to 7 : 5. So which of the two numbers will be larger?
(a) 42 (b) 48
(c) 27 (d) 69
RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (b) Let numbers be 3x, 2x
According to the question $\frac{3x+8}{2x+8} = \frac{7}{5}$
$15x + 40 = 14x + 56$
$x = 16$
now numbers = 3×16, 2×16
= 48, 32 or large number = 48

59. The ratio of red balls and green balls in a bag is 4 : 9. If 7 more red balls are added to the bag, the new ratio of red and green balls will becomes 5 : 6. How many green balls are there in the bag ?
(a) 9 (b) 18
(c) 12 (d) 27
RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (b) Let number of red balls in the bag = 4x
and number of green balls = 9x
Ratio on adding 7 more red balls in the bag 5:6
$\therefore \frac{4x+7}{9x} = \frac{5}{6}$
$24x + 42 = 45x$
$21x = 42$
$x = 2$
$\therefore$ number of green balls in the bag will be
$= 9x = 9 \times 2 = 18$

60. In 9 kg of the dough the initial ratio of sugar and flour is 2 : 7. If John added more sugar in that then the ratio becomes 2 : 5. How much sugar did John added later?
(a) 750 g (b) 1.2 kg
(c) 1kg (d) 800 g
RRB ALP & Tec. (13-08-18 Shift-II)

Ans : (d) Ratio of sugar and flour in dough of 9 kg = 2:7
Then sum of the proportional parts = 2 + 7 = 9
So the quantity sugar $= \frac{2 \times 9}{9} =$ 2 kg
The quantity of flour $= \frac{7 \times 9}{9} = 7$ kg
Suppose that quantity of sugar added later by John is x kg then-
$\frac{2+x}{7} = \frac{2}{5}$
or $10 + 5x = 14$
$5x = 4$
$x = \frac{4}{5} = 0.8$ or $0.8 \times 1000 = 800$ gram

Type - 3

61. The ratio of length and breath of a rectangular board is 7 : 5. If the breath of board is 20.5 cm then, find the length of board.
(a) 19.9 (b) 24.3
(c) 28.7 (d) 14.6
RRB RPF SI -16/01/2019 (Shift-II)

Ans : (c) Let length and breadth of board is 7x and 5x
According to the question
$5x = 20.5$
$x = 4.1$
length = $7x = 7 \times 4.1 = 28.7$ cm.

62. The ratio of the efficiencies of two workers are 3 : 2 . What amount should be given to P out of the daily wages of Rs 400.
(a) Rs. 240 (b) Rs. 260
(c) Rs. 250 (d) Rs. 160
RRB JE - 28/05/2019 (Shift-II)

Ans : (a)

	P		Q
Efficiency	3	:	2

Ratio of efficiency is equal to ratio of wages
Let wages of P = Rs. 3x

Wages of Q= Rs. 2x
Wages of (P + Q) = Rs. 400
3x + 2x = Rs. 400
5x = Rs. 400
x = Rs. 80
Received amount of P = 3x = 3 × 80 = Rs. 240

63. A science book contains two sections of physics and biological science in the ratio of 5 : 4 . How many pieces of biological science will be needed to make a book of 540 pages.
(a) 210 (b) 240
(c) 270 (d) 300

RRB RPF Constable -24/01/2019 (Shift-I)

Ans : (b) Physics : Biological Science = 5 : 4
Number of total pages = 540
Number of pages of biological science $= 540 \times \frac{4}{9}$
= 60 × 4 = 240

64. What is the length of large piece when the 54.6 m long wire is divided in the ratio of 12 : 29.
(a) 15.2 m (b) 7.7 m
(c) 17.4 m (d) 18.4 m

RRB Group-D – 24/09/2018 (Shift-I)

Ans : (c) Length of wire = 24.6 m
Ratio = 12 : 29
Length of large piece $= \frac{24.6 \times 29}{41}$
(∵ 41 = Proportional sum)
= 0.6 × 29 = 17.4 m

65. A person divides Rs. 10,390 into his son and daughter in the ratio of 11: 10. What is the amount received by the son ?
(a) Rs. 5800.32 (b) Rs. 5442.38
(c) Rs. 5390.50 (d) Rs. 6000.68

RRB Group-D – 24/10/2018 (Shift-III)

Ans. (b) : According to the question
Received amount of son $= \frac{11}{11+10} \times 10390$
$= \frac{11}{21} \times 10390 = \frac{114290}{21}$ = Rs. 5442.38

66. An amount is divided between Moksha and Aryan in the ratio of 13 : 29. If Aryan received Rs. 551 then what amount did Moksha get ?
(a) Rs. 234 (b) Rs. 273
(c) Rs. 221 (d) Rs. 247

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (d) Let the received amount of Moksha and Aryan is Rs. 13x and Rs. 29x
According to the question,
29x = 551
x = 19
∴ Received amount of Moksha = 13 ×19 = Rs. 247

67. Rs. 5,100 is distributed between A and B in the ratio of 8 : 9. Share of B is how much more than the share of A.
(a) Rs. 300 (b) Rs. 200
(c) Rs. 250 (d) Rs. 350

RRB Group-D – 23/10/2018 (Shift-I)

Ans. (a) : According to the question,
Ratio of A and B = 8:9
Part of A $= \frac{5100 \times 8}{17} = 2400$
Part of B $= \frac{5100 \times 9}{17} = 2700$
Part of B = Rs. 2700 and Part of A = Rs. 2400
B–A = 2700 – 2400
B–A=Rs.300

68. If 350 marbles are divided between Sita and Savita in the ratio of 5 : 9, then how many marbles will Savita get.
(a) 225 (b) 200
(c) 125 (d) 250

RRB Group-D – 16/10/2018 (Shift-III)

Ans : (a) Let the marbles Sita got = 5x
Marbles Savita got = 9x
9x + 5x = 350
$x = \frac{350}{14} = 25$
Number of marbles Savita got = 9 × 25 = 225

69. The ratio of division of 88 points in Jayati and Pratik is 5 : 6. How much money did Jayati get?
(a) Rs. 40 (b) Rs. 48
(c) Rs. 56 (d) Rs. 52

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (a) Let Jayati got amount = $5x$
Prateek got amount = $6x$
According to the question, $6x + 5x = 88$
$x = 8$
Hence Jayati got amount = $5x = 5 \times 8$ = Rs. 40

70. 77 people are present at a party. The ratio of men and women is 4 : 7. How many women are there in the party ?
(a) 36 (b) 54
(c) 73 (d) 49

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (d) Let the number of men present in the party is 4x and the women is 7x.
Sum of women and men in the party
4x + 7x = 77
11x = 77
x = 7
Of women number = 7x = 7 × 7 = 49

71. The ratio of the number of boys and girls appearing in an examination was 5 : 6. If 3531 students took the exam then how many of them are girls ?

(a) 1936 (b) 1906
(c) 1926 (d) 1916

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (c) According to the question,
Let number of boys = 5x and number of girls = 6x
5x + 6x = 3531
11 x = 3531
$x = \frac{3531}{11} = 321$
Total girls = 6x = 6 × 321 = 1926

72. Rs. 1190 distributed between two persons in the ratio of 11:23. What is the share of the person who is getting the greatest share?
(a) Rs. 828 (b) Rs. 759
(c) Rs. 805 (d) Rs. 782

RRB Group-D – 16/11/2018 (Shift-III)

Ans. (c) : Ratio of rupee distributed between the two persons = 11 : 23
The person receiving the beg part
$= \frac{23}{11+23} \times 1190 \quad = \frac{23}{34} \times 1190 = 23 \times 35 =$ Rs. 805

73. An amount is divided b/w Ram and Shyam in the ratio of 8 : 19. If Shyam received Rs. 247, then how much amount did Ram get ?
(a) Rs. 104 (b) Rs. 112
(c) Rs. 120 (d) Rs. 96

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (a) Let amount of Ram and Shyam is 8x, 19x respectively
From the question
19x = Rs. 247
x = 13
Amount of Ram = 8x = 8 × 13 = Rs. 104

74. The ratio of Salt and acids in a mixture is 5 : 7. If the weight of the mixture is 360g what will be the weight of the acid present in it ?
(a) 260 gm (b) 130 gm
(c) 150 gm (d) 210 gm

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (d) Salt : Acid = 5: 7, Weight of mixture = 360 gm
Weight of acid $= \frac{7}{(5+7)} \times 360 = \frac{7}{12} \times 360$
= 7 × 30 = 210 gm.

75. The ratio of the boys and girls in a school is 4 : 5. If there were 33 students in the school. How many of them were girls
(a) 180 (b) 175
(c) 185 (d) 190

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (c) : Let number of boys = 4x
and number of girls = 5x
number of students = 333
5x + 4x = 333
9 x = 333
x = 37
number of girls = 5x = 37 × 5 = 185

76. In an alloy, the ratio of gold and silver is 17 : 3. If the weight of Silvers in the alloy is 2.7g, find the weight of gold in the alloy.
(a) 12.6 g (b) 15.3 g
(c) 18 g (d) 21.2 g

RRB NTPC 29.03.2016 Shift : 1

Ans :(b) Ratio of gold and Silver in the alloy = 17 : 3
Weight of Silver = 2.7 gram
Weight of gold the alloy $= \frac{17}{3} \times 2.7 = 15.3$ gram

77. One of the 27 nurses in a nursing home has resigned. If the staff ratio is 1 : 6 in proportion to the patient, then how many maximum patients can be admitted.
(a) 156 (b) 162
(c) 150 (d) 168

RRB NTPC 05.04.2016 Shift-1

Ans : (a)
Number of nurses = 27
after resigning number of nurses = 27-1 = 26
∵ nurse : patient = 1:6
so it is clear that 1 nurse = 6 patient
∴ 26 nurse = 26×6 = 156 patient

78. The ratio of science and arts students is 5 : 3. If their total number is 1528 then find the number of students of arts?
(a) 830 (b) 664
(c) 498 (d) 573

RRB NTPC 05.04.2016 Shift-1

Ans : (d) Total number of students = 1528
ratio of student of science and art = 5:3
Hence number of student of art $= \frac{3}{8} \times 1528 = 3 \times 191 = 573$

79. A certain number of men and their dogs are travelling by train. The ratio of men and dogs is 5:2. The train has total 54 legs. Find the total number of dogs.
(a) 30 (b) 12
(c) 15 (d) 6

RRB NTPC 04.04.2016 Shift : 2

Ans : (d) Suppose number of men and dogs are 5x and 2x respectively
from the question-
$5x \times 2 + 2x \times 4 = 54$
$10x + 8x = 54$
$18x = 54$
$x = 3$
So number of dogs = 2x = 2 × 3 = 6

80. The difference b/w two positive numbers is 60, and the ratio of both numbers is 5 : 3. Find the products of both numbers.
(a) 96000 (b) 48000
(c) 144000 (d) 72000

RRB NTPC 04.04.2016 Shift : 3

Ans : (a) Let both positive number is 5x and 3x respectively

$\therefore 5x - 3x = 160$

$2x = 160 \Rightarrow x = 80$

first number = 5x = 5 × 80 = 400

second number = 3x = 3 × 80 = 240

first number × second number = 400 × 240 = 96000

81. If the ratio of two complementary angles is 4 : 5, then find the ratio of the sum of the two angles from their differences.

(a) 1 : 9 (b) 2 : 7
(c) 4 : 9 (d) 7 : 11

RRB NTPC 18.04.2016 Shift : 3

Ans : (a) Let complementary angles is 4x and 5x respectively.

sum of both = 4x + 5x = 9x

difference of both = 5x–4x = x

$\therefore$ Ratio = x : 9x = $\boxed{1:9}$

82. If a 84 m long rope is divided in the ratio 7 : 5, then the length of the longer piece (in m) is:

(a) 42 (b) 50
(c) 45 (d) 49

RRB ALP & Tec. (30-08-18 Shift-III)

Ans : (d) Let the length of piece is = 7x and 5x

$\therefore$ 7x + 5x = 84

12x = 84

x = 7

length of long piece = 7 × 7 = 49 m

83. The ratio of boys and girls in a club is 3 : 2. Which of the following could be the actual number of members?

(a) 18 (b) 25
(c) 16 (d) 24

RRB ALP & Tec. (29-08-18 Shift-I)

Ans : (b) Ratio of boys and girls in club = 3:2

so total probable number = 3+2 = 5

so according to the options total number will be 25

84. If Rs. 87 is divided between James and Radha in the ratio of 1 : 2, how much money did Radha get?

(a) Rs. 29 (b) Rs. 57
(c) Rs. 59 (d) Rs. 58

RRB ALP & Tec. (17-08-18 Shift-I)

Ans : (d) Received amount by Radha

$= \frac{2}{1+2} \times 87 = \frac{2}{3} \times 87 =$ Rs. 58

85. Rs. 69 is divided between Jason and Ruhana in the ratio of 1 : 2. How much money did Ruhana get ?

(a) Rs. 23 (b) Rs. 46
(c) Rs. 40 (d) Rs. 45

RRB ALP & Tec. (14-08-18 Shift-III)

Ans :(b) Suppose Jason got (x) Rs. and Ruhana (2x) Rs.

As per question,

x + 2x = 69

3x = 69

x = 23

Amount of Ruhana = 2x = 2 × 23 = Rs. 46

86. The ratio of the heights of Nani and Leelu is 4 : 3. If Leelu is 1.2 m tall, then what is the height of Nani?

(a) 1.8 m (b) 0.9 m
(c) 2 m (d) 1.6 m

RRB ALP & Tec. (10-08-18 Shift-III)

Ans : (d) Let length of Nani and Lilu is 4x and 3x

According to the question,

3x = 1.2

$\Rightarrow \quad x = \frac{1.2}{3}$

Since length of Nani $4x = 4 \times \frac{1.2}{3}$

$\therefore$ length of Nani $= \frac{4.8}{3} = 1.6$ meter

Type - 5

87. If the ratio of the length of all sides of triangle are in ratio 2 : 3 : 4 and its perimeter is 63 cm, then find the length of longest side.

(a) 30 cm (b) 28 cm
(c) 21 cm (d) 14 cm

RRB RPF SI -13/01/2019 (Shift-II)

Ans : (b) Let sides of triangle is 2x, 3x and 4x

$\therefore$ from the question-

$2x + 3x + 4x = 63$

$9x = 63, \quad x = 7$

$\therefore$ longest side of $\Delta = 4x = 4 \times 7 = 28$ cm.

88. The ratio of the working efficiencies of P, Q, R and S in doing a work is 2 : 3 : 5 : 4. The wages given for task is Rs. 4200, who got the highest amount and how much?

(a) P, Rs. 2000 (b) Q, Rs. 2000
(c) S, Rs. 1600 (d) R, Rs. 1500

RRB JE - 25/05/2019 (Shift-I)

Ans : (d) Total = Rs. 4200

Share of P = $\frac{4200 \times 2}{2+3+5+4} = \frac{4200 \times 2}{14} =$ Rs. 600

Share of Q = $\frac{4200 \times 3}{14} =$ Rs. 900

Share of R = $\frac{4200 \times 5}{14} =$ Rs. 1500

Share of S = $\frac{4200 \times 4}{14} =$ Rs. 1200

So R got the highest amount of Rs. 1500.

89. If Rs. 750 divided in three part x, y and z such a way that (x – 5) : (y – 10) : (z – 15) = 5 : 4 : 3. Then find the share of x, y and z respectively.

(a) Rs. 250, Rs. 200, Rs. 150
(b) Rs. 305, Rs. 250, Rs. 195
(c) Rs. 310, Rs. 170, Rs. 240
(d) Rs. 300, Rs. 240, Rs. 180

RRB JE - 30/05/2019 (Shift-III)

Ans : (b)
(x–5) : (y – 10) : (z–15) = 5 : 4 : 3 = K
x – 5 = 5 K ⇒ x = 5 K + 5 ..... (i)
y – 10 = 4k ⇒ y = 4K + 10 ..... (ii)
z – 15 = 3k ⇒ z = 3K + 15 ..... (iii)
by adding all three equations
(x+y+z) = 12K + 30
750 –30 = 12K
$K = \frac{720}{12}$
K = 60
part of x = 5 K + 5 = 60 × 5 + 5 = 305
part of y = 4 K + 10 = 4 × 60 + 10 = 250
part of z = 3 K + 15 = 3 × 60 + 15 = 195

90. The amount of Rs. 117 is divided between P, Q, R and S by a ratio of 1/2:1/3:1/4 instead of accidentally it divided in another ratio 2:3:4. Who got highest amount in this division and how much does he get?

(a) P, Rs. 28 (b) R, Rs. 27
(c) R, Rs. 25 (d) Q, Rs. 35

RRB JE - 31/05/2019 (Shift-III)

Ans. (c) Actual ratio $P : Q : R = \frac{1}{2} : \frac{1}{3} : \frac{1}{4} = 6 : 4 : 3$
wrong ratio P : Q : R = 2 : 3 : 4
wrong division-

$$\text{part of P} = \frac{2}{9} \times 117 = \text{Rs. } 26$$

$$\text{part of Q} = \frac{3}{9} \times 117 = \text{Rs. } 39$$

$$\text{part of R} = \frac{4}{9} \times 117 = \text{Rs. } 52$$

correct division-

$$\text{part of P} = \frac{6}{13} \times 117 = \text{Rs. } 54$$

$$\text{part of Q} = \frac{4}{13} \times 117 = \text{Rs. } 36$$

$$\text{part of R} = \frac{3}{13} \times 117 = \text{Rs. } 27$$

It is clear that maximum profit gain by R 52 – 27 = Rs. 25 in wrong division.

91. X, Y and Z together complete a work and earn Rs. 210. If the ratio of work of X, Y and Z is 4:6:5 then what is the amount received by Z?

(a) Rs. 40 (b) Rs. 70
(c) Rs. 44 (d) Rs. 45

RRB RPF Constable -25/01/2019 (Shift-I)

Ans : (b) Received amount by Z for work

$$= \frac{5}{4+6+5} \times 210$$

$$= \frac{5}{15} \times 210 = \text{Rs. } 70$$

92. Rs. 900 has been distributed between X, Y and Z in the ratio of 4:5:6. What will be the difference between the amount of X and Z?

(a) 500 (b) 150
(c) 120 (d) 350

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (c)
X : Y : Z = 4 : 5 : 6
distributed amount = Rs. 900

$$\text{difference between X and Z} = \frac{6-4}{15} \times 900$$

$$= \frac{2}{15} \times 900 = 60 \times 2 = \text{Rs. } 120$$

93. 315 marbles are distributed between Aman, Ashok and Alok in the ratio of 7:3:5.

(a) 84 (b) 147
(c) 63 (d) 42

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (a) Number of marbles Aman have

$$= \frac{7}{15} \times 315 = 147$$

number of marbles Ashok have

$$= \frac{3}{15} \times 315 = 21 \times 3 = 63$$

$$\text{number of marbles Alok have} = \frac{5}{15} \times 315 = 105$$

So Aman got marble in compare of Ashok = 147–63 = 84

94. If the angle of triangle are in ratio 5:6:7, then the triangle is called

(a) equilateral triangle
(b) Acute angle triangle
(c) Right triangle
(d) Obtuse angle triangle

Ans : (b) Let the angle of triangle is 5x, 6x, 7x
from the law of triangle
sum of three angles of Δ = 180^0
5x + 6x + 7x = 180^0
18x = 180^0
$x = \frac{180^0}{18} = 10^0$
then angle of Δ, 5x = 50°
6x = 60°
7x = 70°
So it is clear from the three angles of triangle that these all angle is less from 90° so this is the tendency of acute angle.

95. If Rs. 686 is divided into four parts, in ratio of $\frac{1}{2}:\frac{2}{3}:3:4$, then the first part is?

(a) Rs. 48 (b) Rs. 56
(c) Rs. 52 (d) Rs. 42

RRB Group-D – 18/09/2018 (Shift-I)

Ans. (d) : Required Ratio $=\frac{1}{2}:\frac{2}{3}:3:4 = 3:4:18:24$

first part $= \frac{686\times3}{3+4+18+24} = \frac{2058}{49} =$ Rs. 42

96. An amount of Rs. 1000 is distributed between Tara, Tamanna and Teena are in ratio of 4:5:6. What is difference between amount of Tara and Teena?

(a) Rs. 250 (b) Rs. 133.33
(c) Rs. 150 (d) Rs. 234

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (b) : Suppose recieved amount by Tara, Tamanna and Tina is Rs. 4x, Rs. 5x and Rs. 6x respectively.

$\therefore$ $4x + 5x + 6x = 1000$

$15x = 1000$

$x = \frac{200}{3}$

So difference of amount of Tara and Tina

$= x \times (6-4) = 2x$

$= \frac{2\times200}{3} =$ Rs. 133.33

97. Rs. 156 are divided in the ratio of 1:2:4:5. Which of the following order will give the ratio of rupees?

(a) 13,26,53, 64 (b) 13,26,51, 66
(c) 13,26,52, 65 (d) 13,25,53, 65

RRB NTPC 22.04.2016 Shift : 2

Ans : (c) Ratio of rupees-

$\Rightarrow \frac{1}{1+2+4+5}\times156 = \frac{1}{12}\times156 = 13$

$\Rightarrow \frac{2}{12}\times156 = 26$

$\Rightarrow \frac{4}{12}\times156 = 52$

$\Rightarrow \frac{5}{12}\times156 = 65$

Intended answer $\Rightarrow$ 13, 26, 52, 65

98. If Rs. 130 are divided in ratio 3:6:8:9 what amount will be given to each other.

(a) 15, 30, 41, 44 (b) 15, 30, 40, 45
(c) 15, 30, 39, 46 (d) 15, 31, 39, 45

RRB NTPC 17.01.2017 Shift-3

Ans : (b) Sum of proportion = 3+6+8+9 = 26

first part $= \frac{3}{26}\times130 = 15$

second part $= \frac{6}{26}\times130 = 30$

third part $= \frac{8}{26}\times130 = 40$

fourth part $= \frac{9}{26}\times130 = 45$

99. A certain prize money is given to P, Q, R and S in ratio 3:5:7. If R gets Rs. 1000 more than P, what is share of Q?

(a) Rs. 750 (b) Rs. 1250
(c) Rs. 1750 (d) Rs. 1500

RRB NTPC 05.04.2016 Shift-1

Ans : (b) Let given reward amount of P, Q and R is 3x, 5x and 7x respectively.

According to the question $7x - 3x = 1000$

$\Rightarrow 4x = 1000$

$\Rightarrow x = 1000/4 = 250$

$\therefore$ share of Q $= 5x = 5\times250 =$ Rs.1250

100. If there are three numbers in the ratio of 2:5:7 and their sum is half of the 280, then what is the square of the smallest number among the three?

(a) 400 (b) 900
(c) 2500 (d) 6400

RRB NTPC 03.04.2016 Shift : 3

Ans : (a) Let three numbers are 2x, 5x and 7x respectively

According to the question,

$2x + 5x + 7x = \frac{280}{2}$

$14x = 140$, $\boxed{x = 10}$

The square of smallest number $= (2x)^2 = (2\times10)^2 = (20)^2 = 400$

101. If the three numbers are in the ratio of 4:5:7 and their sum is 320. Find the sum of the smallest and the largest number among them.

(a) 140 (b) 220
(c) 240 (d) 180

Ans (b) Let numbers are 4x, 5x and 7x.

$\therefore$ $4x + 5x + 7x = 320$

$16x = 320$

$x = 20$

$\therefore$ required sum $= 4x + 7x = 11x = 11\times20 = 220$

102. If three number are in the ratio of 1:3:5 and their sum is 10800 then find the largest number in three numbers.

(a) 1200 (b) 3600
(c) 6000 (d) 5400

RRB NTPC 02.04.2016 Shift : 3

Ans : (c)

First number $= \frac{1}{1+3+5}\times10,800 = \frac{1}{9}\times10,800 = 1200$

second number $= \frac{3}{9}\times10,800 = 3600$

third number $= \frac{5}{9}\times10800 = 5\times1200 = 6000$

so largest number = 6000

103. If the ratio of angles of a quadrilateral is 3:5:9:13 then find the largest angle?

(a) 165° (b) 180°
(c) 156° (d) 190°

RRB NTPC 29.03.2016 Shift : 1

Ans : (c) Ratio of angles of quadrilateral $= 3:5:9:13$
Let angles of quadrilateral are 3x, 5x, 9x and 13x
$\because$ sum of four angles of quadrilateral $= 360°$
$\therefore \quad 3x + 5x + 9x + 13x = 360°$
$30x = 360°$ or $x = 12°$
largest angle of quadrilateral $= 13x = 13 \times 12 = 156°$

104. If the ratio of the angles of a triangle is 2:3:7, find the ratio of the largest angle to the smallest angle.

(a) 7:2 (b) 2:3
(c) 7:1 (d) 3:5

RRB NTPC 18.04.2016 Shift : 1

Ans : (a) Let the angles of triangle are 2x, 3x and 7x respectively.
$\therefore 2x + 3x + 7x = 180°$
$12x = 180°$
$\Rightarrow x = \frac{180°}{12} \quad x = 15°$
so largest angle $= 7x$
$= 7 \times 15° = 105°$
Smallest angle $= 2x$
$= 2 \times 15° = 30°$
Ratio $= 105 : 30 = 7:2$

105. If the ratio of the angles of a triangle is 1:2:3, then find the smallest angles.

(a) 30° (b) 60°
(c) 70° (d) 45°

RRB NTPC 18.04.2016 Shift : 3

Ans : (a) Let the angles of triangle are x, 2x and 3x respectively.
According to the question,
$x + 2x + 3x = 180°$ {sum of three angles of triangle is 180°}
$\Rightarrow 6x = 180°$
$\Rightarrow x = 30°$
So smallest angle $= x = 30°$

106. If the ratio of the angles of a triangle is 1:4:7 then find the ratio of the smallest angle to the largest angle.

(a) 7 : 2 (b) 2 : 3
(c) 7 : 1 (d) 3 : 5

RRB NTPC 12.04.2016 Shift : 3

Ans : (c) sum of three angles of triangle is 180°
$\therefore$ smallest angle $= \frac{1}{1+4+7} \times 180 = \frac{1}{12} \times 180 = 15°$
largest angle $= \frac{7}{1+4+7} \times 180° = \frac{7}{12} \times 180° = 105°$
required ratio $= 105 : 15 = 7 : 1$

107. If Rs. 176 is divided in the ratio of 2:5:6:9 the rupees will be.

(a) 16, 40, 49 and 71 (b) 16, 40, 47 and 73
(c) 16, 40, 48 and 72 (d) 16, 41, 47 and 72

RRB NTPC 07.04.2016 Shift : 1

Ans : (c) Let the ratio $= 2x : 5x : 6x : 9x$
$\Rightarrow 2x + 5x + 6x + 9x = 176$
$22x = 176$
$x = \frac{176}{22}$
$x = 8$
first part $= 2 \times 8 = 16$
second part $= 5 \times 8 = 40$
third part $= 6 \times 8 = 48$
fourth part $= 9 \times 8 = 72$

108. Ritesh, Gaurav and Jugnu invest in the ratio of 2:5:7. The ratio of income after return of their investment is 5:3:2. If Gaurav earns more than Rs. 150 from Ritesh, find the total income.

(a) Rs. 1170 (b) Rs. 1050
(c) Rs. 1240 (d) Rs. 1370

RRB NTPC 22.04.2016 Shift : 1

Ans : (a) Let investment of Ritesh, Gaurav and Jugnu are 2x, 5x and 7x respectively.
Ratio of income after returning of investment
$= \frac{2x \times 5}{100} : \frac{5x \times 3}{100} : \frac{7x \times 2}{100}$
$= \frac{10x}{100} : \frac{15x}{100} : \frac{14x}{100}$
According to the question-
$\frac{15x}{100} - \frac{10x}{100} = 150$
$\Rightarrow \frac{5x}{100} = 150$
$\Rightarrow x = 3000$
total income $= \frac{10 \times 3000}{100} + \frac{15 \times 3000}{100} + \frac{14 \times 3000}{100}$
$= 300 + 450 + 420 =$ Rs. 1170

109. Mother distributed some amount in her Children, Arvind, Benu and Chitra in the ratio of 2:3:4. If Beenu receive Rs. 51, then what will be the amount received by Chitra and Arvind respectively?

(a) 34, 102 (b) 85, 34
(c) 68, 34 (d) 68, 136

RRB NTPC 28.04.2016 Shift : 3

Ans : (c) Let the amount of Arvind, Beenu and Chitra are 2x, 3x and 4x respectively.
$\therefore$ 3x (amount of Beenu) = 51
x = Rs. 17
$\therefore$ received amount by Chitra and Arvind = 4x and 2x
$= 4 \times 17$ and 2×17
= 68 and 34

Type - 3

110. If 12, x, 48 are in consecutive ratio, then find the value of x.

(a) 24 (b) 21
(c) 36 (d) 16

RRB RPF Constable -19/01/2019 (Shift-III)

Ans : (a)
$\because$ 12, x and 48 are in consecutive ratio
$\therefore$ 12 : x : : x : 48
$\Rightarrow \frac{12}{x} = \frac{x}{48}$
$x^2 = 12 \times 48$
$x^2 = \overline{2\times2} \times 3 \times \overline{2\times2} \times \overline{2\times2} \times 3$
$x = 2 \times 2 \times 2 \times 3 = 24$

111. In x : 1/18 : : 3/5 : 5/9 find the value of x.

(a) 4/25 (b) 3/50
(c) 3/25 (d) 7/9

RRB RPF SI -06/01/2019 (Shift-II)

Ans : (b) $x : \frac{1}{18} :: \frac{3}{5} : \frac{5}{9}$
$x \times \frac{5}{9} = \frac{1}{18} \times \frac{3}{5}$
$x = \frac{1}{18} \times \frac{3}{5} \times \frac{9}{5}$
So, $x = \frac{3}{50}$

112. The first three terms of any ratio are 3, 5 and 21 respectively. Find its fourth term.

(a) 25 (b) 35
(c) 30 (d) 20

RRB RPF Constable -22/01/2019 (Shift-I)

Ans. (b) : Let fourth term is x
3 : 5 :: 21: x
$\frac{3}{5} = \frac{21}{x}$
$3x = 21 \times 5$
$x = \frac{21 \times 5}{3} = 7 \times 5 = 35$

113. Which of the following should be added in each to make the numbers 4, 8, 12, 22 proportional?

(a) $\frac{4}{3}$ (b) $\frac{3}{4}$
(c) $\frac{8}{3}$ (d) $\frac{5}{6}$

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (a) Let that number is k
then,
4 + k : 8 + k : : 12 + k : 22 + k
$\frac{4+k}{8+k} = \frac{12+k}{22+k}$
$88 + 4k + 22k + k^2 = 96 + 12k + 8k + k^2$
$88 + 26k = 96 + 20k$
$6k = 8$
$k = \frac{8}{6} = \frac{4}{3}$
So $k = \frac{4}{3}$ should be added in each number.

114. If 25, 35 and P are consecutive ratio, find the value of p.

(a) 60 (b) 50
(c) 75 (d) 49

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (d) $\because$ 25, 35 and P are in consecutive ratio
So, $\frac{25}{35} = \frac{35}{P}$
$P = \frac{35 \times 35}{25}$
$P = 7 \times 7 = 49$

115. Find the fourth proportional of 3, 4 and 9 ____

(a) 11 (b) 12
(c) 16 (d) 10

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (b) Let fourth proportion at is x
then, 3 : 4 :: 9 : x
$\frac{3}{4} = \frac{9}{x}$
$3x = 9 \times 4$
$x = \frac{9 \times 4}{3}$
x = 12

116. If 5 : 9 : : x : 27 then what is value of x ?

(a) 6 (b) 3
(c) 18 (d) 15

RRB Paramedical Exam – 21/07/2018 (Shift-II)

Ans : (d) 5 : 9 : : x : 27
$\frac{5}{9} = \frac{x}{27}$
$\boxed{x = 15}$

117. If p : 18 : : 5 : 3, then find the value of p

(a) 60 (b) 30
(c) 25 (d) 50

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (b) : Proportion-
p : 18 : : 5 : 3
$\therefore$ $p \times 3 = 18 \times 5$
$p = \frac{18 \times 5}{3}$
$p = 6 \times 5$
p = 30

118. If 7 : 9 : : x : 36 then find the value of x.

(a) 21 (b) 35
(c) 7 (d) 28

RRB Group-D – 16/11/2018 (Shift-I)

Ans. (d)
7 : 9 : : x : 36
$9x = 36 \times 7$
$9x = 252$
x = 28

119. Find the fourth proportional number of 2, 4 and 8.

(a) 15 (b) 14
(c) 16 (d) 18

RRB NTPC 17.01.2017 Shift-3

Ans : (c) Let fourth proportional number = x

$\therefore \quad 2 : 4 :: 8 : x$

$2 \times x = 4 \times 8$ (by proportional law)

$x = \frac{4 \times 8}{2} \Rightarrow \boxed{x = 16}$

120. In $\frac{26}{21} : \frac{24}{9} :: k : \frac{14}{13}$ find the value of k.

(a) $\frac{1}{3}$ (b) 2
(c) $\frac{1}{2}$ (d) 3

RRB NTPC 18.01.2017 Shift : 1

Ans : (c) $\frac{26}{21} : \frac{24}{9} :: k : \frac{14}{13}$

$\Rightarrow \frac{26}{21} \times \frac{14}{13} = \frac{24}{9} \times K$

$\Rightarrow K = \frac{26 \times 14 \times 9}{21 \times 13 \times 24} \qquad \Rightarrow K = \frac{1}{2}$

121. Find the value of x only if the three numbers are 2.6, 1.3 and x are in certain ratio.

(a) 1.95 (b) 1.83
(c) 3.9 (d) 0.65

RRB NTPC 18.01.2017 Shift : 2

Ans : (d) 2.6 : 1.3 : : 1.3 : x

$\frac{2.6}{1.3} = \frac{1.3}{x}$

$x = \frac{1.3 \times 1.3}{2.6} \qquad \boxed{x = 0.65}$

122. 4 terms are in consecutive ratio first, second and fourth terms are 4, 22 and 33. Then find the third term.

(a) 8 (b) 6
(c) 11 (d) 3

RRB NTPC 18.01.2017 Shift : 2

Ans : (b) Let third term = P

$\Rightarrow 4 : 22 :: P : 33$

$\Rightarrow 4 \times 33 = 22 \times P$

$P = \frac{4 \times 33}{22} = 6$

123. Find the fourth proportional of 9, 17 and 27?

(a) 57 (b) 48
(c) 51 (d) 53

RRB NTPC 06.04.2016 Shift : 1

Ans : (c) Let fourth proportional of 9, 17 and 27 is x

$\therefore 9 : 17 :: 27 : x$

$9 \times x = 17 \times 27 \Rightarrow x = \frac{17 \times 27}{9}$

$\Rightarrow x = 17 \times 3 = 51$

124. What is the fourth proportional number of 14, 28 and 21?

(a) 41 (b) 42
(c) 14 (d) 40

RRB NTPC 19.01.2017 Shift : 1

Ans : (b) 14 : 28 : : 21 : x

fourth proportional number $(x) = \frac{28 \times 21}{14} = 42$

125. What is the fourth proportional 15, 12, 20 ?

(a) 14 (b) 12
(c) 18 (d) 16

RRB NTPC 19.01.2017 Shift : 2

Ans : (d) Let fourth proportion = P

$\Rightarrow 15 : 12 :: 20 : P$

$\Rightarrow \frac{15}{12} = \frac{20}{P}$

$\Rightarrow 15 \times P = 12 \times 20$

Hence $\boxed{P = 16}$

126. If 14, x and 56 are in consecutive ratio, find the value of x.

(a) 28 (b) 21
(c) 8 (d) 42

RRB NTPC 30.04.2016 Shift : 1

Ans : (a)

$\because$ 14, x and 56 is in are in consecutive ratio

$\therefore 14 : x :: x : 56$

$x^2 = \sqrt{14 \times 56}$

$x^2 = \sqrt{7 \times 2 \times 2 \times 2 \times 2 \times 7}$

$x = 7 \times 2 \times 2 = 28$

127. The mean proportional between 0.16 and 0.64 is:

(a) 0.27 (b) 0.48
(c) 0.40 (d) 0.32

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (d) Let mean proportional between 0.16 and 0.64 is x.

$\therefore 0.16 : x :: x : 0.64$

According to the question, $\frac{0.16}{x} = \frac{x}{0.64}$

$x^2 = 0.16 \times 0.64$

$x = 0.4 \times 0.8$

$x = 0.32$

128. Find the mean proportional between 2 and 98?

(a) 13 (b) 14.5
(c) 16 (d) 14

RRB ALP & Tec. (10-08-18 Shift-II)

Ans : (d) Then proportion between 2 and 98

$= \sqrt{2 \times 98} = \sqrt{2 \times 7 \times 14} = \sqrt{14 \times 14} = 14$

Type - 3

129. There are a total of 221 rupees in the form of 50 paise, Rs. 1 and Rs. 2 in Manu's Piggy bank whose ratio is 4:3:6 what is number of 50 paise coins in Piggy bank?

(a) 52 (b) 13
(c) 104 (d) 26

RRB RPF Constable -17/01/2019 (Shift-I)

Ans : (a) Let number of coins of 50 paise 1 rupees and 2 rupees are 4x, 3x and 6x
According to the question,

$$\frac{4x}{2}+3x+2\times 6x=221$$

17x = 221 x = 13

Number of coins of 50 paise = 4x = 4 × 13 = 52

130. In a bag 25 paise, 10 paise and 5 paise coins are in the ratio 1:2:3. If the bag has Rs. 30 in total then how many coins of 5 paise are there?

(a) 100 (b) 200
(c) 150 (d) 50

RRB JE - 22/05/2019 (Shift-I)

Ans : (c) Let number of coins of 25 paise, 10 paise and 5 paise are x, 2x and 3x respectively.
25 × x + 10 × 2x + 5 × 3x = 3000
25x + 20x + 15x = 3000
60x = 3000
x = 50
So number of coins of 5 paise = 3x = 3 × 50 = 150

131. A boy has total amount Rs. 60 in which the coin of 50 paise and 25 paise are included in the ratio of 5:6:8. Find the no. of coins of 25 paise?

(a) 30 (b) 48
(c) 32 (d) 42

RRB RPF SI -12/01/2019 (Shift-II)

Ans : (b) Let the number of coins of Rs. 1, 50 paise and 25 paise is 5x, 6x and 8x respectively

value of coins ⇒ Rs. $\left(\frac{5x}{1}+\frac{6x}{2}+\frac{8x}{4}\right)$

⇒ Rs. (5x + 3x + 2x)
= 10x
but total cost = Rs. 60 (given)
∴ 10x = 60
x = 6
so number of coins of 25 paise = 8 × 6 = 48

132. In a bag, the coin of 50 paise, 25 paise and 10 paise are in the ratio of 5:4:3. If the value of coin is Rs. 171. Find the number of all types of coins.

(a) 200, 250, 150 (b) 225, 180, 135
(c) 140, 150, 280 (d) 200, 360, 160

RRB Group-D – 15/11/2018 (Shift-I)

Ans : (b) Let number of coins of 50 paise, 25 paise and 10 paise in the bag is 5x, 4x, 3x respectively

value of coins = $\frac{5x}{2}:\frac{4x}{4}:\frac{3x}{10}$

According to the question,

⇒ $\frac{5x}{2}+\frac{4x}{4}+\frac{3x}{10}=171$

⇒ 50x + 20x + 6x = 3420
⇒ 76x = 3420,
⇒ x = 45

Hence no of coins = 5x = 5 × 45 = 225
= 4x = 4 × 45 = 180
= 3x = 3 × 45 = 135

i.e. coins of 50 paise, 25 paise and 10 paise is 225, 180, 135 respectively.

133. The ratio of the numbers of blue and red balls in the bag is constant. When there were 44 red balls, the number of blue balls was 36. If the number of blue balls is 54, then what will be the number of red balls?

(a) 66 (b) 62
(c) 64 (d) 68

RRB ALP & Tec. (09-08-18 Shift-I)

Ans : (a) Let number of red balls = x
According to the question,

$$\frac{44}{36}=\frac{x}{54}$$

$$x=\frac{44\times 54}{36}$$

x = 66

Hence number of red balls will be 66 in the bag.

134. Suraj have the coin of 50 paise, Rs. 1 and Rs. 5 in the ratio of 8:5:9. Suraj have total amount of Rs. 648. How many coins of 50 paise does he has?

(a) 96 (b) 84
(c) 60 (d) 108

RRB Group-D – 11/12/2018 (Shift-I)

Ans. (a) :
Suppose number of coins of 50 paise, Rupees 1, Rupees 5 is 8x, 5x and 9x respectively

then total value of these = Rs. $\left(\frac{8x}{2}+5x\times 1+9x\times 5\right)$

= Rs. $(4x+5x+45x)$ = Rs. 54x

∴ 54x = 648
x = 12

number of coins of 50 paise = 8x = 8×12 = 96

135. Ramesh have the coins of 50 paise, Rs. 1 and Rs. 5 in the ratio of 2:3:5 respectively. He has a total amount of Rs. 116. How many coins of 50 paise he has?

(a) 12 (b) 6
(c) 4 (d) 8

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (d) :
Suppose number of coins of 50 paise Rs, 1, Rs 5 is 2x, 3x, 5x respectively

then total value of these $= \frac{1}{2}\times 2x + 1\times 3x + 5\times 5x$

According to the question,

$x + 3x + 25x = 116$

$29x = 116$

$x = 4$

Hence number of coins of 50 paise $= 2x = 2 \times 4 = 8$ coins

136. Raju has Rs. 210 as coins 20% of the coins are Rs. 5, 25% Rs. 10 and 15% is Rs. 2 and remaining Rs. 1 are in the denominations. Find the no. of coins of Rs. 1.

(a) 24 (b) 22
(c) 25 (d) 20

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (d) Raju has a total of Rs. 210 as coins
Let the total ratio of the number of coins of Raju is 100%

Rs. 5	Rs.10	Rs. 2	Rs. 1
20%	25%	15%	40%

Ratio of number of coins $= \frac{1}{5}:\frac{1}{4}:\frac{3}{20}:\frac{2}{5}$

$= 4 : 5 : 3 : 8$

Let the number of coins $= 4x, 5x, 3x, 8x$

Cost of total number of coins =

$4x \times 5 + 5x \times 10 + 3x \times 2 + 8x \times 1 = 210$

$20x + 50x + 6x + 8x = 210$

$84x = 210$

$x = \frac{210}{84} = \frac{10}{4}$

Hence number of coins of Rs. 1 $= 8x = 8\times\frac{10}{4} = 20$

137. An amount of Rs. 110 is in the form of coins of Rs. 2, Rs. 1 and 50 paise, which is the ratio of 1:2:3 respectively. How many coins of 50 paise are in it.

(a) 20 (b) 40
(c) 60 (d) 80

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (c) Let number of coins of Rs. 2, Rs. 1 and 50 Paise is x, 2x and 3x respectively.
Total amount = Rs. 110
According to the question,

$(x\times 2) + (2x \times 1) + (3x \times 0.50) = 110$

$\Rightarrow 2x + 2x + 1.5x = 110$

$\Rightarrow 5.5\,x = 110$

$\Rightarrow x = \frac{110}{5.5} = 20$

$\Rightarrow x = 20$

Number of coins of 50 paise $= 3x = 3 \times 20 = 60$

138. In a bag of Rs. 10, Rs. 20 and Rs. 50 notes are kept in the ratio of 1:3:5 if the total of notes amount kept in the bag is Rs. 1920. What is the total number of Rs. 20?

(a) 6 (b) 30
(c) 18 (d) 12

RRB NTPC 18.01.2017 Shift : 2

Ans : (c)
Let number of notes of Rs. 10, Rs, 20, Rs. 50 = x, 3x, 5x
According to the question,
Total cost $= 10 \times x + 20 \times 3x + 50 \times 5x = 1920$

$\Rightarrow 320x = 1920$

$\Rightarrow x = 6$

Total number of notes of 20 Rs. = 3x

$= 3 \times 6 = 18$

139. In a bag coins of 25 paise, 10 paise and 5 paise are in ratio 1:2:3. If their total value is Rs. 180, then how many coins of 10 paise are found in it?

(a) 300 (b) 400
(c) 600 (d) 900

RRB NTPC 09.04.2016 Shift : 3

Ans : (c) Let ratio of coins of 25 paise, 10 paise and 5 paise is x, 2x and 3x.

25 paise = Rs. $\frac{1}{4}$

10 paise = Rs. $\frac{1}{10}$

5 paise = Rs. $\frac{1}{20}$

$\frac{1}{4}x + \frac{2}{10}x + \frac{3}{20}x = 180$

$\frac{5x + 4x + 3x}{20} = 180$

$12x = 180 \times 20$

$x = \frac{180\times 20}{12}$

$x = 15\times 20$ or $x = 300$

Number of coins of 10 paise $2x = 300\times 2 = 600$

140. The value of coins of Rs. 1, 50 paise and 25 paise is Rs. 93.75 and the ratio of their number is 3:4:5. Find the no. of coins of each type.

(a) 42, 56, 70 (b) 45, 60, 75
(c) 40, 70, 75 (d) 46, 58, 75

RRB ALP & Tec. (17-08-18 Shift-II)

Ans : (b) Let number of Rs. coins of 1, 50 paise and 25 paise is 3x, 4x and 5x respectively
According to the question,

$3x + \frac{4x}{2} + \frac{5x}{4} = 93.75$

$\Rightarrow \frac{12x + 8x + 5x}{4} = 93.75$

$25x = 93.75\times 4$

$x = \frac{375}{25} \Rightarrow x = 15$

number of coins of Rs. 1 $=3\times 15=45$
number of coins of 50 paise $=4\times 15=60$
number of coins of 25 paise $=5\times 15=75$

141. A bag contains coins of Rs. 1, Rs. 5 and Rs. 10. Assuming that there are equal number of coins of Rs. 1, Rs. 5 and Rs. 10 what will the total number of coins in the bag if the total money is Rs. 368 ?

(a) 69 (b) 96
(c) 56 (d) 65

RRB ALP & Tec. (14-08-18 Shift-III)

Ans : (a) Let the coins of all three types are in equal number (x) then
there are coins of 1 Rs.
there are coins of 5 Rs.
there are coins of 10 Rs.
According to the question,

$$x + 5x + 10x = 368$$
$$16x = 368$$
$$x = 23$$

number of total coins $= 3x = 3 \times 23 = 69$

Type - 8

142. Divide Rs. 305 into three parts in this way the 3/7 first of the, 2/3 of the second, 4/4 of the third are in equal find the third part.

(a) Rs. 62 (b) Rs. 72
(c) Rs. 75 (d) Rs. 40

RRB RPF SI -16/01/2019 (Shift-II)

Ans : (c) Let the three parts be x, y and z Rs. respectively

$$\therefore \frac{3}{7}x = \frac{2}{3}y = \frac{4}{5}z = k \text{ (Suppose)}$$

$$\therefore x = \frac{7k}{3}, y = \frac{3k}{2}, z = \frac{5k}{4}$$

$$x : y : z = \frac{7k}{3} : \frac{3k}{2} : \frac{5k}{4} = 28 : 18 : 15$$

$$\text{part of third } (z) = 305 \times \frac{15}{(28+18+15)}$$

$$= 305 \times \frac{15}{61} = \text{Rs. } 75$$

143. Divide 1870 in three part. So that half of the first part 1/3 of the second part and 1/6 of the third part are in equal. What will be the all three parts ?

(a) 360, 510, 1000
(b) 340, 510, 1020
(c) 340, 490, 1040
(d) 360, 490, 1020

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (b) According to the question,
Half of first part = 1/3 of second part = 1/6 of part third

$$\frac{I}{2} = \frac{II}{3} = \frac{III}{6}$$

I : II : III = 2 : 3 : 6

$$\text{I part} = 1870 \times \frac{2}{11} = 340$$

$$\text{II part} = 1870 \times \frac{3}{11} = 510$$

$$\text{III part} = 1870 \times \frac{6}{11} = 1020$$

Hence all three parts are 340, 510 and 1020

144. 1380 divided between A, B and C in which A gets 5 times of C and 3 times of B-

(a) A = 880; B = 200; C = 300
(b) A = 800; B = 300; C = 280
(c) A = 980; B = 300; C = 100
(d) A = 900; B = 300; C = 180

RRB Group-D – 01/10/2018 (Shift-III)

Ans : (d) According to the question,

$$A = 5C, \ A = 3B$$
$$\Rightarrow \quad A = 3B = 5C$$
$$\Rightarrow \quad \frac{A}{15} = \frac{B}{5} = \frac{C}{3}$$

A : B : C = 15 : 5 : 3
proportional sum = 23

$$A = \frac{1380 \times 15}{23} = \text{Rs. } 900$$

$$B = \frac{1380 \times 5}{23} = \text{Rs. } 300$$

$$C = \frac{1380 \times 3}{23} = \text{Rs. } 180$$

145. Rs. 13,680 is divide into three part like that first part is 3/5 of third part, second and third part are in ratio 4:7, How much will be the first part?

(a) 3780 (b) 6300
(c) 1600 (d) 4800

RRB NTPC 29.03.2016 Shift : 1

Ans : (a) Let first, second and third part is x, y and z respectively.

$$\therefore x = \frac{3}{5}z \Rightarrow x : z = 3 : 5$$

$$y : z = 4 : 7 \Rightarrow z : y = 7 : 4$$

x	z	y
3	5	
	7	4
21 :	35 :	20

$$\text{first part} = \frac{21}{76} \times 13680 = 3780$$